

Project Initialization and Planning Phase

Date	01 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	3 Marks

Define Problem Statements :

Loan approval is one of the most critical and frequent decisions made by banks and non-banking financial companies (NBFCs). Traditionally, loan officers manually review an applicant's income, credit history, employment status, and other financial details before approving or rejecting a loan. This manual process is time-consuming, inconsistent across officers, and prone to human bias, which can result in delayed decisions for applicants and increased default risk for lenders.

This project addresses that gap by building a machine learning-powered loan eligibility prediction system trained on historical loan application data. Using classification algorithms — Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost — the system analyses applicant attributes such as income, marital status, education, credit history, and property area to predict whether a loan application should be approved or rejected.

The best-performing model is saved and integrated into a Flask web application, enabling loan officers and applicants to receive a real-time, data-driven eligibility prediction along with a confidence score and risk classification through a simple and accessible web interface.